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Fractions tricks — Numbers 69 to 80

Each passage is now a mini-lesson: a challenge to hook them in, the trick, one worked example, two for the child to try, then both checked together — with encouragement all the way through. Read it like you're at the kitchen table and someone just asked "go on then, show me".

The four rules

1. Before each passage say its number clearly — "Number one", "Number two" — pause a second, read the passage, then pause a couple of seconds before the next number. Those gaps are how I find the joins. 2. Wherever you see ... leave a short beat — two seconds is plenty. Those are the "have a think" moments for the kids. 3. If you stumble, don't stop the recording: pause, say the SAME number again, and read that passage again. I'll keep your retake. 4. Kitchen-table warmth beats newsreader polish every time.

Total is roughly two hours of reading — absolutely fine to do over several sittings (each take just needs to start at a numbered item; tick them off on the printed version as you go). When done, drop the audio file(s) — m4a, mp3, wav, anything — into OneDrive Times tables/recordings and tell me.

Tick each number as you record it so you can pick up where you left off.

Part E — Fractions tricks

■ Say: "Number 69" — Same Bottom → Add the Tops

Quick one — what's $\frac{3}{8}$ plus $\frac{2}{8}$? ... If you got that in a flash, you already know today's rule. If it made you freeze, don't worry — this is the friendliest fraction rule there is. When two fractions have the same bottom number, you leave the bottom completely alone and just add — or subtract — the tops. The bottom tells you the size of the pieces; the tops tell you how many pieces. Let's do $\frac{3}{8}$ plus $\frac{2}{8}$. The bottoms match — both are 8 — so the bottom stays 8. Add the tops: 3 plus 2 is 5. So $\frac{3}{8}$ plus $\frac{2}{8}$ is $\frac{5}{8}$. Done — no clever tricks needed. Your turn — pause me if you need to: what's $\frac{4}{9}$ plus $\frac{3}{9}$? ... and what's $\frac{6}{11}$ minus $\frac{2}{11}$? ... Have a proper go before I tell you... Let's check together. $\frac{4}{9}$ plus $\frac{3}{9}$: the bottoms match, so keep the 9. Add the tops: 4 plus 3 is 7. The answer is $\frac{7}{9}$. Now $\frac{6}{11}$ minus $\frac{2}{11}$: bottoms match, keep the 11. Subtract the tops: 6 minus 2 is 4. The answer is $\frac{4}{11}$. Same bottom, sort the tops — that's the whole rule. Everything else in fractions builds on this one, so you're off to a flying start.

■ Say: "Number 70" — Magic Fraction Facts

Quick fire — what's $\frac{1}{4}$ as a percentage? ... If you said 25 percent before I finished asking, you've already banked this one. If not, brilliant — today is about collecting a handful of magic facts. A few fraction facts, learned by heart, make almost every fraction, decimal and percentage question instant. Here are the stars: $\frac{1}{2}$ is 0.5, which is 50 percent. $\frac{1}{4}$ is 0.25, which is 25 percent. $\frac{3}{4}$ is 0.75, which is 75 percent. $\frac{1}{5}$ is 0.2, which is 20 percent. $\frac{1}{10}$ is 0.1, which is 10 percent. So for our opener: $\frac{1}{4}$ is 0.25 — and 0.25 means 25 out of 100, which is 25 percent. No calculating, just knowing. Your turn — pause me if you need to: what's $\frac{3}{4}$ as a percentage? ... and what's $\frac{1}{5}$ as a decimal? ... Have a proper think before I tell you... Let's check. $\frac{3}{4}$ is 0.75, so as a percentage it's 75 percent — three lots of 25. And $\frac{1}{5}$ is 0.2 as a decimal, which is 20 percent. Keep saying these facts until they pop out without thinking — then you'll barely have to calculate at all. That's a trick for life, that one.

■ Say: "Number 71" — Fraction of a Number

Quick one — what's $\frac{3}{4}$ of 20? ... If you nailed that in under ten seconds, you're already flying. If not, no bother — this trick works for any fraction of any number. To find a fraction of a number, divide by the bottom, then multiply by the top. Always divide first — it keeps the numbers nice and small. Let's do $\frac{3}{4}$ of 20. The bottom is 4, so divide: 20 divided by 4 is 5. The top is 3, so multiply: 5 times 3 is 15. So $\frac{3}{4}$ of 20 is 15. Divide by the bottom, times by the top — every single time. Your turn — pause me if you need to: what's $\frac{2}{5}$ of 30? ... and what's $\frac{3}{8}$ of 40? ... Have a proper go before I tell you... Right, let's check together. $\frac{2}{5}$ of 30: divide by the bottom — 30 divided by 5 is 6. Multiply by the top — 6 times 2 is 12. So $\frac{2}{5}$ of 30 is 12. Now $\frac{3}{8}$ of 40: divide by the bottom — 40 divided by 8 is 5. Multiply by the top — 5 times 3 is 15. So $\frac{3}{8}$ of 40 is 15. See? Bottom first, then top — you've got this one sorted.

■ Say: "Number 72" — Simplify by Halving

Quick one — can you write $\frac{12}{16}$ as a simpler fraction? ... If you got there in under ten seconds, you clearly know your halves. If not, perfect timing — here's the easiest simplifying trick there is. If the top and the bottom of a fraction are both even, halve them both. Then look again — if they're still both even, halve again. Keep going until you can't any more. Let's do $\frac{12}{16}$. Both even? Yes. Halve both: 12 becomes 6, and 16 becomes 8, giving $\frac{6}{8}$. Both still even, so halve again: 6 becomes 3, and 8 becomes 4, giving $\frac{3}{4}$. Now 3 is odd, so we stop. $\frac{12}{16}$ fully simplified is $\frac{3}{4}$. Your turn — pause me if you need to: simplify $\frac{8}{20}$... and simplify $\frac{24}{32}$... Have a proper go before I tell you... Let's check together. $\frac{8}{20}$: both even, so halve both to get $\frac{4}{10}$. Both still even — halve again to get $\frac{2}{5}$. The 5 is odd, so we're done: $\frac{8}{20}$ is $\frac{2}{5}$. Now $\frac{24}{32}$: halve both to get $\frac{12}{16}$... halve again to get $\frac{6}{8}$... and once more to get $\frac{3}{4}$. So $\frac{24}{32}$ is $\frac{3}{4}$. Halving first, every time — see, you've got this one.

■ Say: "Number 73" — Slash the Zeros

Quick one — can you make 30 over 50 simpler? ... If you saw the answer in a flash, you already know this trick. If not, no worries at all — it takes about a minute to learn. Here's the secret: if the top and the bottom of a fraction both end in 0, you can slash a zero off each one. That's really just dividing the top and bottom by 10, so the fraction stays worth exactly the same. Let's do 30 over 50. The top ends in 0, the bottom ends in 0 — slash them both off. 30 becomes 3, and 50 becomes 5. So 30 over 50 is simply 3 over 5. Done in one stroke of the pencil. Your turn — pause me if you need to: simplify 40 over 90 ... and simplify 500 over 800 ... Have a proper go before I tell you. Ready? 40 over 90: both end in 0, so slash the zeros — that's 4 over 9. And 500 over 800: slash one zero from each — 50 over 80. They still both end in 0, so slash again — 5 over 8. See? You've got this one — go slash some zeros in the practice.

■ Say: "Number 74" — Compare to One Half

Quick challenge — is 5 over 8 more than a half, or less than a half? ... If you knew that straight away, brilliant — you already own this trick. If you had to guess, stick with me for one minute. Here's the idea: a fraction is exactly a half when the top number is exactly half the bottom number. So just ask — what's half of the bottom? If the top is bigger than that, the fraction is more than a half. If it's smaller, it's less. Let's check 5 over 8. Half of 8 is 4. Is 5 more than 4? Yes — so 5 over 8 is more than a half. No sums, just a quick look. Your turn — pause me if you need to: is 3 over 10 more or less than a half? ... and is 7 over 12 more or less than a half? ... Have a proper think before I tell you. Right — 3 over 10: half of 10 is 5, and 3 is less than 5, so 3 over 10 is less than a half. And 7 over 12: half of 12 is 6, and 7 is more than 6, so 7 over 12 is more than a half. That's it — you can now size up a fraction at a glance.

■ Say: "Number 75" — Cross-Multiply to Compare

Quick one — which is bigger, $\frac{2}{3}$ or $\frac{5}{8}$? ... If you cracked that in under ten seconds, hats off — you know this already. If not, you're about to learn a proper ninja move. To compare two fractions, cross-multiply: each top times the OTHER fraction's bottom. Whichever side makes the bigger product wins. It works because you're secretly making a common denominator without writing it down. Let's settle $\frac{2}{3}$ against $\frac{5}{8}$. Take 2 times 8 — that's 16. Now 5 times 3 — that's 15. 16 is bigger, and the 16 came from the $\frac{2}{3}$ side — so $\frac{2}{3}$ is the bigger fraction. Your turn — pause me if you need to: which is bigger, $\frac{3}{4}$ or $\frac{5}{7}$? ... and which is bigger, $\frac{2}{5}$ or $\frac{3}{7}$? ... Have a proper go before I spill the answers. Here we go. $\frac{3}{4}$ against $\frac{5}{7}$: 3 times 7 is 21, and 5 times 4 is 20. 21 wins, and it came from $\frac{3}{4}$ — so $\frac{3}{4}$ is bigger. Next, $\frac{2}{5}$ against $\frac{3}{7}$: 2 times 7 is 14, and 3 times 5 is 15. 15 wins, so $\frac{3}{7}$ is bigger. See? You've got this one — go smash the practice.

■ Say: "Number 76" — Cancel Before Multiplying

Quick one — what's $\frac{3}{4}$ times $\frac{8}{9}$? ... If that felt easy, you already know today's trick. If it felt scary, don't worry — this is the trick that makes it easy. When you multiply fractions, look for shared factors and slash them out FIRST, before you multiply. You can cancel any top with any bottom — diagonally or in the same fraction — but never top with top. Let's do $\frac{3}{4}$ times $\frac{8}{9}$. The 3 on top and the 9 on the bottom share a 3 — cancel them down to 1 and 3. The 8 on top and the 4 on the bottom share a 4 — cancel them down to 2 and 1. Now it's just 1 over 1 times 2 over 3, which is $\frac{2}{3}$. Tiny numbers, no mess. Your turn — pause me if you need to: what's $\frac{2}{3}$ times $\frac{9}{10}$? ... and what's $\frac{5}{6}$ times $\frac{3}{10}$? ... Have a proper go first. Right — $\frac{2}{3}$ times $\frac{9}{10}$: the 2 and the 10 cancel to 1 and 5, the 9 and the 3 cancel to 3 and 1, leaving $\frac{3}{5}$. And $\frac{5}{6}$ times $\frac{3}{10}$: the 5 and the 10 cancel to 1 and 2, the 3 and the 6 cancel to 1 and 2, leaving 1 over 2 times 1 over 2 — that's $\frac{1}{4}$. That's it — cancel first, and multiplying fractions becomes a doddle.

■ Say: "Number 77" — Keep, Change, Flip

Quick challenge — what's a half divided by a quarter? ... If you said 2 without blinking, you already know this famous trick. If dividing fractions makes your brain wobble, perfect — this fixes it forever. The trick is called Keep, Change, Flip. KEEP the first fraction exactly as it is. CHANGE the divide sign into a times sign. FLIP the second fraction upside down. Then just multiply. Let's do 1 over 2 divided by 1 over 4. Keep the 1 over 2. Change divide to times. Flip 1 over 4 into 4 over 1. Now multiply: 1 over 2 times 4 over 1 is 4 over 2, and 4 over 2 is 2. So a half divided by a quarter is 2 — because two quarters fit inside a half. Your turn — pause me if you need to: what's 3 over 4 divided by 1 over 8? ... and what's 1 over 3 divided by 1 over 6? ... Have a proper go before I tell you. Ready? 3 over 4 divided by 1 over 8: keep 3 over 4, change to times, flip to 8 over 1. That's 24 over 4, which is 6. And 1 over 3 divided by 1 over 6: keep 1 over 3, times, flip to 6 over 1 — that's 6 over 3, which is 2. See? Keep, change, flip — you now know a trick most grown-ups have forgotten.

■ Say: "Number 78" — Butterfly Method

Quick one — what's 2 over 3 plus 1 over 4? ... If you got 11 over 12 in under ten seconds, you're flying — you know this one. If not, let me show you the butterfly. When two fractions have different bottoms, draw a butterfly: multiply across the wings to get the tops, and multiply the two bottoms for the new bottom. No hunting for a common denominator — the butterfly does it for you. Let's do 2 over 3 plus 1 over 4. First wing: 2 times 4 is 8. Second wing: 1 times 3 is 3. Add those for the top: 8 plus 3 is 11. Now the bottoms: 3 times 4 is 12. So 2 over 3 plus 1 over 4 is 11 over 12. Your turn — pause me if you need to: what's 1 over 2 plus 1 over 5? ... and what's 3 over 5 plus 1 over 4? ... Have a proper go before I tell you. Here we go. 1 over 2 plus 1 over 5: wings first — 1 times 5 is 5, and 1 times 2 is 2. Top: 5 plus 2 is 7. Bottom: 2 times 5 is 10. Answer: 7 over 10. And 3 over 5 plus 1 over 4: 3 times 4 is 12, 1 times 5 is 5, top is 12 plus 5 — 17. Bottom: 5 times 4 is 20. Answer: 17 over 20. That's it — flap those wings and go smash the practice.

■ Say: "Number 79" — Spot the Easy Common Bottom

Quick one — what's 1 over 4 plus 3 over 8? ... If you spotted the shortcut and got 5 over 8, you're already ahead. If not, here's the little check that saves loads of work. Before you reach for any fancy method, look at the two bottoms and ask: is one a multiple of the other? If it is, you only need to change ONE fraction, and the sum becomes easy. Let's do 1 over 4 plus 3 over 8. Is 8 a multiple of 4? Yes — 4 times 2 is 8. So change 1 over 4 into eighths: multiply top and bottom by 2, and 1 over 4 becomes 2 over 8. Now the bottoms match: 2 over 8 plus 3 over 8 is 5 over 8. Done. Your turn — pause me if you need to: what's 1 over 3 plus 2 over 9? ... and what's 1 over 2 plus 3 over 8? ... Have a proper go before I tell you. Right then. 1 over 3 plus 2 over 9: 9 is a multiple of 3, so 1 over 3 becomes 3 over 9. Then 3 over 9 plus 2 over 9 is 5 over 9. And 1 over 2 plus 3 over 8: 8 is a multiple of 2, so 1 over 2 becomes 4 over 8. Then 4 over 8 plus 3 over 8 is 7 over 8. See? Always hunt for the easy common bottom first — the butterfly is just the back-up plan.

■ Say: "Number 80" — Fraction → Percent

Quick challenge — what's 3 over 5 as a percentage? ... If you said 60 percent before I finished asking, you already own this trick. If not, one minute from now you will. Per cent just means out of 100. So to turn a fraction into a percentage, make the bottom into 100 — multiply the top and the bottom by the same number until the bottom hits 100. Whatever lands on top is your percentage. Let's do 3 over 5. What turns 5 into 100? Multiply by 20. So multiply the top by 20 as well: 3 times 20 is 60. That gives 60 over 100 — which is exactly 60 percent. So 3 over 5 is 60 percent. Your turn — pause me if you need to: what's 7 over 20 as a percentage? ... and what's 9 over 25 as a percentage? ... Have a proper go before I tell you. Ready? 7 over 20: to get from 20 to 100, multiply by 5. So 7 times 5 is 35 — that's 35 over 100, which is 35 percent. And 9 over 25: 25 times 4 is 100, so 9 times 4 is 36 — that's 36 over 100, which is 36 percent. That's it — you can now turn fractions into percentages faster than a calculator comes out of the drawer.

Done with this part? Save the audio into OneDrive > Times tables > recordings. Thank you!